Problem

Problem · A miniature RSA cipher. Choose two primes $p = 5$ and $q = 11$, and let $n = pq = 55$ and $\varphi = (p-1)(q-1) = 40$.

1. Choose $e = 3$. Explain why e and φ are coprime, then find an integer $d \in \{1, \dots, 39\}$ such that $ed \equiv 1 \pmod{40}$.
2. Encryption: a message is an integer $m \in \{0, \dots, 54\}$; it is encrypted as the remainder c of m^e modulo n. Encrypt $m = 8$.
3. Decryption: we take for granted for now that $c^d \equiv m \pmod{n}$. Check with a calculator or Python that $17^{27} \equiv 8 \pmod{55}$.
4. We prove that $m^{ed} \equiv m \pmod{55}$ for every integer m.
   (a) Write $ed = 1 + 40k$ with k a natural number.
   (b) If $5 \nmid m$, prove that $m^{40k} \equiv 1 \pmod{5}$; deduce $m^{ed} \equiv m \pmod{5}$. What happens if $5 \mid m$?
   (c) Establish likewise $m^{ed} \equiv m \pmod{11}$, then conclude.
5. Why does the security of this cipher rest on the difficulty of factorising n when p and q are very large?